

Template for Computer Programming I Practical Recording

Date	: 23/01/2023
Problem Specification	: Calculate the interest for the month, the total new balance (previous balance + additional charges + interest), and the minimum payment due of the credit cards.
Assumption	: The user inputs are valid.
Limitation	: N/A
Input	: Input the previous balance and additional charges of the credit card.
Processing	: the program read the previous balance, additional charges. then Calculate the interest of the previous total balance and find the minimum payment due of the credit cards.
Output	: print the previous balance, additional charge, interest, minimum payment due of the credit cards.
Algorithm	<pre>:Step1 :Prompt "Enter the previous balance" Step2: Initialize variable previousBalance for previous balance Step3: Read the previousBalance given by user Step4: Assign input value for previousBalance Step5:Prompt "Enter the additional charges" Step6: Initialize variable additionalCharges for additional Charges. Step7: Read the input additionalCharges given by user Step8: Assign input value for additionalCharges Step9: Print previous balance as "Previousbalance is:\$" Step10: Print additionalCharges as "Additional Charges: \$" Step11: Initialize variable : previousTotalBalance ← previousBalance+additionalCharges; Step12: if(previousTotalBalance >0) is true Initialize variable : interest ← previousTotalBalance*2/100; Step13 : Print interest as "Interest:\$" Step14: Initialize variable : newBalance ← previousTotalBalance+interest; Step15: Print as newBalance as "New balance:\$"- Step16:if (newBalance <+50) is True Print newBalance as "Minimum payment:\$ " else if(newBalance >50 && newBalance <300) is True Print as "Minimum payment: \$50" else Initialize variable : newPayment ← newBalance*20/100; Print newPayment as "Minimum payment: \$"</pre>

Step17: if (newBalance <+50) is False
Print "the previous balance was 0 "
Step18: End the program

Program listing : Programme file **attached**

Test data and expected output : **1. Test data:** Enter the previous balance: 15000
Enter the additional charges: 2000

Expected output: Previous balance is: \$15000.0
Additional Charges: \$2000.0
Interest: \$340.0
New balance: \$17340.0
Minimum payment: \$3468.0

Output obtained for test data : **1. Test data:** Enter the previous balance: 15000
Enter the additional charges: 2000

Obtained output: Previous balance is: \$15000.0
Additional Charges: \$2000.0
Interest: \$340.0
New balance: \$17340.0
Minimum payment: \$3468.0

Discussion : The implementation was successful in meeting the requirements

Conclusion : The program accurately calculates the Interest, New balance and Minimum payment.